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Consider the following objects:

$$q.5 \quad s = \begin{bmatrix} 140 \\ 90 \\ 75 \\ 110 \end{bmatrix} \quad C = \begin{bmatrix} 4 & 7 & 5 \\ 6 & 3 & 8 \end{bmatrix}$$

$$X_{pijt}, \text{ where } \begin{aligned} p &= 1, \dots, 3 \\ i &= 1, \dots, 2 \\ j &= 1, \dots, 4 \\ t &= 1, \dots, 5. \end{aligned}$$

a) Classify each object as a scalar, vector, matrix, or tensor. Give the dimensions.

$q.5 \rightarrow$ scalar. Dimensions $1 \times 1 \quad \mathbb{R}$

$s \rightarrow$ vector. Dimensions $4 \times 1 \quad \mathbb{R}^4$

$C \rightarrow$ Matrix. Dimensions $2 \times 3 \quad \mathbb{R}^{2 \times 3}$

$X \rightarrow$ Tensor. Dimensions $3 \times 2 \times 4 \times 5$

b) Give one logistics interpretation for s , C , and X_{pijt} . (Each)

s : Vector 4×1

$\Rightarrow$ stock levels of a single product stored across 4 warehouses.

Example:

WH1 $\Rightarrow$ 140 units

WH2 $\Rightarrow$ 90 units

WH3 $\Rightarrow$ 75 units

WH4 $\Rightarrow$ 110 units

Matrix C : (2×3)

$\Rightarrow$ In the logistics interpretation where columns are sources and rows are targets, C can be a representation of the cost to ship a pallet of goods from 3 production facilities to 2 distribution centers.

Example:

$C_{12} = 7$, means it costs 7 EUR to ship from facility 2 to distribution center 1.

X_{pijt} : Tensor $3 \times 2 \times 4 \times 5$

$\Rightarrow$ Spatio-temporal supply chain flow volume.

Example:

p	product type	1 to 3
i	target retailer	1 to 2
j	source supplier	1 to 4
t	time (days)	1 to 5

$\Rightarrow$ The value at $X_{3,1,4,2}$ gives us the # units of product 3 that were shipped to retailer 1 from supplier 4 on day 2.

c) Determine the total scalar entries contained in S , C , and X .

S : $4 \times 1 \Rightarrow 4$ entries

C : $2 \times 3 \Rightarrow 6$ entries

X : $3 \times 2 \times 4 \times 5 \Rightarrow 120$ entries

Total: 130 entries

d) Interpret the statement in a precise logistics sentence:

$$X_{2,1,3,4} = 18$$

$\Rightarrow$ "18 units of product 2 were shipped from supplier 3 to retailer 1 on day 4."

3) Three Suppliers provide quantities

$$q = \begin{bmatrix} 50 \\ 40 \\ 60 \end{bmatrix}.$$

The allocation matrix from suppliers to 2 hubs is:

$$A = \begin{bmatrix} 0.6 & 0.3 & 0.2 \\ 0.4 & 0.7 & 0.8 \end{bmatrix}.$$

Rows represent hubs and columns suppliers.

a) State the dimensions of A , q , and Aq

$$A: 2 \times 3$$

$$q: 3 \times 1$$

$$Aq: (2 \times 3) \cdot (3 \times 1) = 2 \times 1$$

b) Compute the hub in-flow vector

$$h = Aq.$$

$$h = \begin{bmatrix} 0.6 & 0.3 & 0.2 \\ 0.4 & 0.7 & 0.8 \end{bmatrix} \begin{bmatrix} 50 \\ 40 \\ 60 \end{bmatrix} = \begin{bmatrix} 0.6(50) + 0.3(40) + 0.2(60) \\ 0.4(50) + 0.7(40) + 0.8(60) \end{bmatrix}$$
$$= \begin{bmatrix} 54 \\ 96 \end{bmatrix}$$

c) Check the column sums of A . What do they imply about conservation of total quantity?

$$A = \begin{bmatrix} 0.6 & 0.3 & 0.2 \\ 0.4 & 0.7 & 0.8 \end{bmatrix}$$

Supplier 1

$$0.6 + 0.4 = 1.0$$

Supplier 2

$$0.3 + 0.7 = 1.0$$

Supplier 3

$$0.2 + 0.8 = 1.0$$

$\Rightarrow$ All columns sum to 1.0, implying a strict conservation of total quantity.

$\Rightarrow$ No loss!

Supplier quantity

$$\begin{bmatrix} 50 \\ 40 \\ 60 \end{bmatrix}$$



150

Hub quantity

$$\begin{bmatrix} 54 \\ 96 \end{bmatrix}$$



150

$\longleftrightarrow$
same

d) Interpret the entry $a_{23} = 0.8$ in one logistics sentence.

$\Rightarrow$ "80% of the total quantity provided by supplier 3 is allocated to hub 2."

7 A logistics planner must choose the numbers of 3 vehicle (truck) types.

$x = \text{small}$

$y = \text{medium}$

$z = \text{large}$

The planning constraints are

$$2x + 4y + 8z = 68$$

pallet capacity

$$x + y + z = 15$$

number of vehicles

$$x + 2y + 3z = 30$$

driver-hours

a) Write the system in matrix:

$$Ax = b$$

$\Rightarrow$ Recognize that the constraint equations describe the "consumption" of resources.

We have 3 decision variables and 3 "consumable" resources.

Dimensional Analysis

$$\frac{2 \text{ pallet}}{1 \text{ s. vehicle}} | x \text{ s. vehicles} + \frac{4 \text{ pallets}}{1 \text{ m. vehicle}} | y \text{ m. vehicles} + \frac{8 \text{ pallets}}{1 \text{ L. vehicle}} | z \text{ L. vehicles} = 68 \text{ pallets}$$

$$1 | x \text{ s. vehicles} + 1 | y \text{ m. vehicles} + 1 | z \text{ L. vehicles} = 15 \text{ vehicles}$$

$$\frac{1 \text{ hrs}}{1 \text{ s. vehicle}} | x \text{ s. vehicles} + \frac{2 \text{ hrs}}{1 \text{ m. vehicle}} | y \text{ m. vehicles} + \frac{3 \text{ hrs}}{1 \text{ L. vehicle}} | z \text{ L. vehicles} = 30 \text{ hrs.}$$

Matrix form:

$$A = \begin{bmatrix} 2 & 4 & 8 \\ 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}, \quad x = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad b = \begin{bmatrix} 68 \\ 15 \\ 30 \end{bmatrix}$$

$$Ax = b \Rightarrow \begin{bmatrix} 2 & 4 & 8 \\ 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 68 \\ 15 \\ 30 \end{bmatrix}$$

b) Solve the system using Gaussian elimination.

⇒ Objective: Get an augmented matrix in row-echelon form.

$$\left[\begin{array}{ccc|c} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array} \right]$$

Strategy: We have 3 row operations we can use.

1. Swap any 2 rows.
2. Multiply any row by a non-zero constant
3. Add a multiple of one row to another row.

4 Steps:

1. Choose a pivot (leading non-zero entry).
2. Use row operations to eliminate entries from rows below the pivot.
3. Work from top to bottom, left to right.
4. Back substitute

⇒ Our problem: Augmented Matrix

$$\begin{array}{l} R_1 \\ R_2 \\ R_3 \end{array} \left[\begin{array}{ccc|c} 2 & 4 & 8 & 68 \\ 1 & 1 & 1 & 15 \\ 1 & 2 & 3 & 30 \end{array} \right]$$

1. R_1 is the pivot.

$$2. \quad R_2 \leftarrow R_2 - R_3 \quad \Rightarrow \quad 0 \quad -1 \quad -2 \quad | \quad -15$$

$$R_3 \leftarrow R_3 - \frac{1}{2} R_1 \quad \Rightarrow \quad 0 \quad 0 \quad -1 \quad | \quad -4$$

$$3. \quad \left[\begin{array}{ccc|c} 2 & 4 & 8 & 68 \\ 0 & -1 & -2 & -15 \\ 0 & 0 & -1 & -4 \end{array} \right]$$

$$\begin{array}{rcl}
 4. & 2x + 4y + 8z = 68 & (1) \\
 & -y - 2z = -15 & (2) \\
 & -z = -4 & (3)
 \end{array}$$

$$\begin{array}{l}
 (3) \Rightarrow z = 4 \\
 \downarrow \\
 (2) \quad -y - 2(4) = -15 \\
 \quad \quad -y = -7 \Rightarrow y = 7 \\
 \downarrow \\
 (1) \quad 2x + 4(7) + 8(4) = 68 \\
 \quad \quad 2x + 28 + 32 = 68 \\
 \quad \quad 2x + 60 = 68 \\
 \quad \quad 2x = 8 \\
 \quad \quad x = 4
 \end{array}$$

Solved: $x = 4$, $y = 7$, $z = 4$

$\swarrow$ small vehicles $\downarrow$ med. vehicles $\searrow$ Large vehicles

c) see (b)

d) Explain what a negative or non-integer solution would mean in a logistics planning context.

$\Rightarrow$ A negative solution would be meaningless – we can't have negative vehicles.

$\Rightarrow$ A non-integer solution is meaningless until rounded. But then care has to be taken to account for the context of the problem.

Ex: Round up or down?